

Lab 3 - Architecture Selection Justification

Multi-Cloud Asset Cost Optimization Portal

Name	SRN	Section
R Vinay	PES1UG24CS353	F

Architecture Selection

We chose Microservices Architecture for the Multi-Cloud Asset Cost Optimization Portal.

Reason 1 - Independent Scalability

The portal performs different workloads such as billing-data ingestion, cost classification, idle and unattached asset detection, and budget monitoring. Separating these functions into independent services allows individual services to be scaled according to their workload.

Reason 2 - Independent Deployment and Fault Isolation

The portal integrates with AWS, Azure, and GCP. Microservices allow cloud-provider integration and other functional services to be developed, updated, and deployed independently. A failure in one service is less likely to affect the entire portal.

Security Advantage

Each service can enforce authentication and authorization at its boundary. This is particularly important for asset-resolution operations because the Cloud Administrator must have permission to act on the relevant cloud account or project.

Performance Benefit

Resource-intensive operations such as processing billing data and scanning cloud assets can be scaled independently, improving performance when workloads increase.

Basis for Selection: The Lab 3 reference identifies independent scaling, fault isolation, and independent deployment as key advantages of Microservices Architecture. The selection also matches the portal functions established in Lab 1, including billing ingestion, cost categorization, idle/unattached asset detection, budget alerts, and asset resolution.